Anemia

All key points for the exam + drug summary table

1. Definition

Population	Anemia if Hgb is
Adult men	**< 13 g/dL**
Adult non-pregnant women	**< 12 g/dL**
Pregnant women	**< 11 g/dL** (1st and 3rd trimesters), < 10.5 g/dL (2nd trimester)

Anemia is a **sign**, not a diagnosis: always look for the **underlying cause**.

2. Signs and Symptoms

Type	Signs and symptoms
General (all anemias)	**Fatigue**, weakness, **pallor**, **dyspnea on exertion**, **tachycardia**, dizziness, headache, cold intolerance, chest pain (in CAD)
Iron deficiency	**Pica** (craving ice = pagophagia, clay, starch), **koilonychia** (spoon nails), **glossitis**, **angular cheilitis**, restless legs, hair loss
Vitamin B12 deficiency	**Neurologic symptoms**: **paresthesias** (numbness, tingling), **loss of vibration and position sense**, **ataxia**, **memory loss, dementia, depression, psychosis** • Glossitis (beefy red tongue) • Neuro damage may be **irreversible**
Folate deficiency	Same as B12 hematologically, but **NO neurologic symptoms**
Hemolysis	**Jaundice**, **dark urine**, splenomegaly

Key difference: B12 deficiency → neurologic symptoms. Folate deficiency → no neurologic symptoms.

3. Classification by MCV (red cell size)

Type	MCV	Causes
Microcytic	**< 80 fL**	**Iron deficiency** (most common) • **Thalassemia** • Anemia of chronic disease (late) • **Sideroblastic** anemia • **Lead** poisoning **Mnemonic: TAILS** (Thalassemia, Anemia of chronic disease, Iron deficiency, Lead, Sideroblastic)
Normocytic	**80–100 fL**	**Anemia of chronic disease/inflammation** • **CKD** • **Acute blood loss** • **Hemolysis** • **Aplastic anemia** • Mixed deficiencies (iron + B12)
Macrocytic	**> 100 fL**	**Megaloblastic: vitamin B12** or **folate** deficiency, **drugs** • **Non-megaloblastic: alcohol**, liver disease, **hypothyroidism**, reticulocytosis, myelodysplastic syndrome

Drugs causing macrocytosis (high-yield)

- **Antifolates: methotrexate**, **trimethoprim**, **pyrimethamine**, pemetrexed
- **Antimetabolites / chemo: hydroxyurea** (expected effect), azathioprine, 6-mercaptopurine, 5-fluorouracil, capecitabine
- **Antiretrovirals: zidovudine** (AZT)
- **Anticonvulsants: phenytoin**, phenobarbital, primidone, valproate
- **B12 absorption reducers: metformin**, **PPIs**, **H2RAs**, colchicine, **nitrous oxide**
- Others: sulfasalazine, oral contraceptives (minor)

4. Key Laboratory Tests

Test	Meaning
Hgb / Hct	Severity of anemia
MCV	RBC size (microcytic / normocytic / macrocytic)
MCH / MCHC	Hemoglobin content (low = **hypochromic**, e.g., iron deficiency)
RDW (red cell distribution width)	Variation in RBC size • **High in iron deficiency** • **Normal in thalassemia**
Reticulocyte count	Bone marrow response • **Low** → production problem (deficiency, marrow failure) • **High** → **blood loss** or **hemolysis**, or response to treatment
Serum iron	Circulating iron
TIBC (total iron-binding capacity)	Reflects transferrin • **Rises** when the body needs iron
TSAT (transferrin saturation)	Serum iron ÷ TIBC × 100 • Iron available for erythropoiesis
Ferritin	**Iron stores** (best single test for iron deficiency) • **Acute phase reactant**: falsely normal/high in inflammation
Vitamin B12 (cobalamin)	Low (< 200 pg/mL) in B12 deficiency
Methylmalonic acid (MMA)	**High in B12 deficiency only** (normal in folate deficiency)
Homocysteine	**High in BOTH** B12 and folate deficiency
Peripheral smear	**Hypersegmented neutrophils** (megaloblastic) • **Schistocytes** (MAHA) • **Heinz bodies, bite cells** (G6PD) • **Target cells** (thalassemia) • **Sickle cells**
Hemolysis labs	↑ **LDH**, ↑ **indirect bilirubin**, ↓ **haptoglobin**, ↑ **reticulocytes** • **Direct antiglobulin test (DAT, Coombs)**: positive in immune hemolysis

Differentiating microcytic anemias

Test	Iron deficiency	Anemia of chronic disease	Thalassemia trait
Serum iron	↓	↓	Normal or ↑
TIBC	↑	**↓ or normal**	Normal
TSAT	↓ (< 20%)	↓ or normal	Normal or ↑
Ferritin	↓ (< 30; < 15 is diagnostic)	**Normal or ↑**	Normal or ↑
RDW	↑	Normal	**Normal**
RBC count	↓	↓	**Normal or ↑** (Mentzer index MCV/RBC **< 13**)

Differentiating B12 vs folate deficiency

Test	B12 deficiency	Folate deficiency
MMA	↑	**Normal**
Homocysteine	↑	↑
Neurologic symptoms	**Yes**	**No**
Intrinsic factor / parietal cell antibodies	Positive in **pernicious anemia**	Negative

5. Iron Deficiency Anemia (IDA)

Causes

Category	Examples
Blood loss (most common in adults)	**GI bleeding** (ulcers, NSAIDs, cancer), **heavy menstrual bleeding** • In men and postmenopausal women, **rule out GI malignancy**
Increased requirements	**Pregnancy**, lactation, infancy, adolescence (growth)
Decreased intake	Vegetarian/vegan diet, poor diet, elderly, infants fed **cow's milk before 12 months**
Decreased absorption	**Celiac disease**, **gastric bypass/bariatric surgery**, **H. pylori**, atrophic gastritis, **PPIs/H2RAs** (iron needs acid)

Iron requirements (RDA)

Group	Elemental iron/day
Adult men, postmenopausal women	**8 mg**
Premenopausal women	**18 mg**
Pregnancy	**27 mg** (supplement with ~**30–60 mg** elemental iron daily, more if anemic)

Iron is absorbed in the **duodenum** as **ferrous (Fe^{2+})**; **acid** and **vitamin C** increase absorption. **Hepcidin** (from the liver) blocks iron absorption and release; it rises after an iron dose and with inflammation.

Oral iron (first-line)

Product	Elemental iron
Ferrous sulfate 325 mg	**65 mg (20%)**
Ferrous sulfate, exsiccated (dried) 200 mg	**~65 mg (30%)**
Ferrous gluconate 325 mg	**~36–38 mg (12%)**
Ferrous fumarate 324 mg	**~106 mg (33%)**
Polysaccharide-iron complex, **carbonyl iron**	**100%** elemental (dose stated as elemental iron)
Ferric maltol (Accrufer)	30 mg twice daily (newer, better GI tolerance)

Dosing

- **Adults: 60–200 mg elemental iron/day** (classic: ferrous sulfate 325 mg 1–3 times daily).
- **Once-daily or alternate-day dosing** is often as effective with **fewer GI effects** (because of **hepcidin** rise after each dose).
- **Children: 3–6 mg/kg/day** elemental iron.
- Higher doses do **not** increase absorption much, but **increase GI side effects**.

Administration and counseling

- Take on an **empty stomach** (1 hour before or 2 hours after meals) for best absorption.
- May take **with food** if GI upset (absorption decreases by up to ~50%, but improves adherence).
- **Vitamin C** (or orange juice) increases absorption.
- **Avoid** taking with **tea, coffee, milk/dairy, calcium, antacids, high-fiber foods**.
- **Liquid iron** can **stain teeth** → use a straw or rinse the mouth.
- **Black/dark stools** are **expected** (harmless; must be differentiated from melena).
- Keep away from **children**: iron overdose is a leading cause of fatal poisoning in young children.
- **Enteric-coated / sustained-release** products have **less GI upset but poorer absorption** (iron released past the duodenum).

Adverse effects

Constipation, nausea, epigastric pain, diarrhea, **dark stools**, metallic taste, teeth staining (liquid).

Drug interactions (high-yield)

Drug	Interaction
Antacids, PPIs, H2RAs	↓ Iron absorption (iron needs acid)
Calcium, dairy, phosphate binders	↓ Iron absorption → separate
Tetracyclines (doxycycline, minocycline)	**Chelation** → ↓ absorption of **both** → separate
Fluoroquinolones (ciprofloxacin, levofloxacin)	**Chelation** → take quinolone **2 h before or 6 h after** iron
Levothyroxine	↓ Levothyroxine absorption → separate by **4 hours**
Integrase inhibitors (dolutegravir, bictegravir, raltegravir)	Chelation → take INSTI 2 h before or 6 h after (or together with food for some)
Levodopa, methyldopa, penicillamine	↓ Absorption of these drugs
Bisphosphonates	↓ Absorption
Mycophenolate	↓ Absorption
Cefdinir	↓ Absorption + **red/reddish stools**
Vitamin C	↑ Iron absorption

Expected response (monitoring)

Time	Expected response
3–4 days	Reticulocytes start to rise
7–10 days	**Peak reticulocytosis**
2–4 weeks	**Hgb rises ≥ 1–2 g/dL** (if not → evaluate)
~6–8 weeks (2 months)	Hgb normalizes
Continue ~3 months after Hgb normalizes	To **replete iron stores** (ferritin)

Reasons for oral iron failure

1. **Nonadherence** (GI side effects) — most common
2. **Ongoing blood loss**
3. **Malabsorption** (celiac, bariatric surgery, PPIs, taking with food/interacting drugs)
4. **Wrong diagnosis** (e.g., thalassemia, anemia of chronic disease)
5. Concurrent deficiency (B12, folate) or inflammation

IV (parenteral) iron

Indications

- **Intolerance** or **failure** of oral iron
- **Malabsorption** (IBD, celiac, **bariatric surgery**)
- **Ongoing blood loss** greater than oral iron can replace
- **CKD on hemodialysis** (and patients receiving **ESAs**)
- Need for **rapid repletion** (e.g., late pregnancy, before surgery)
- Chemotherapy-induced anemia with ESA

Total iron deficit (Ganzoni formula)

Iron deficit (mg) = Weight (kg) × [Target Hgb − Actual Hgb (g/dL)] × **2.4** + **iron stores** (**500 mg** if weight ≥ 35 kg)
Iron in blood: **1 mL of packed RBCs ≈ 1 mg of iron**.

IV iron products

Product	Usual dose	Key points
Iron dextran (INFeD)	**Test dose 25 mg** (wait ≥ 1 h), then calculated dose (can give total dose infusion)	**Boxed warning: anaphylaxis** (fatal reactions) • **Test dose required** • LMW dextran only
Sodium ferric gluconate (Ferrlecit)	**125 mg** IV per HD session × **8** doses (1 g)	No test dose • Hypotension
Iron sucrose (Venofer)	HD: **100 mg × 10** doses (1 g) • Non-dialysis CKD: **200 mg × 5** doses over 14 days	No test dose • Hypotension, cramps
Ferumoxytol (Feraheme)	**510 mg** IV, then **510 mg 3–8 days later**	**Boxed warning: anaphylaxis** • Infuse over **≥ 15 min**, monitor **30 min** • **Interferes with MRI** (up to 3 months)
Ferric carboxymaltose (Injectafer)	**750 mg × 2** doses ≥ 7 days apart (≥ 50 kg) • 15 mg/kg if < 50 kg	**Hypophosphatemia** (can be severe, osteomalacia) • Hypertension, flushing
Ferric derisomaltose (Monoferric)	**1,000 mg single dose** (≥ 50 kg) • 20 mg/kg if < 50 kg	Hypersensitivity • Less hypophosphatemia

- **General IV iron ADRs: hypersensitivity/anaphylaxis**, **hypotension**, flushing, arthralgia/myalgia, iron overload.
- **Monitor for hypersensitivity** for at least **30 minutes** after infusion; have resuscitation equipment available.
- **Stop oral iron** while giving IV iron; **do not give during active infection**.
- **Hold** if ferritin > 500–800 ng/mL or TSAT > 50% (iron overload).
- Check iron studies **≥ 4 weeks** after the last IV dose (earlier results are falsely high).

Iron overload and toxicity

Situation	Treatment
Acute iron poisoning (children, overdose)	**Deferoxamine IV** (up to **15 mg/kg/h**) • **"Vin rosé" (reddish) urine** during therapy • Activated charcoal does **not** bind iron
Chronic iron overload (repeated transfusions: thalassemia, sickle cell, MDS)	**Chelation:** deferoxamine, **deferasirox**, **deferiprone** (start when ferritin > 1,000 ng/mL or after ~20 units of RBCs)

Chelator	Dose	Key ADRs / notes
Deferoxamine (Desferal)	**20–60 mg/kg/day SC** over 8–12 h, 5–7 days/week (chronic)	Injection site reactions • **Ototoxicity, visual (retinal) toxicity** • **Yersinia** infections • Red urine
Deferasirox (Exjade: dispersible tablet; Jadenu: tablet)	Exjade **20 mg/kg/day** (empty stomach) • Jadenu **14 mg/kg/day**	**Boxed warning: renal failure, hepatic failure, GI hemorrhage** • Avoid **aluminum antacids** • ↑ **repaglinide** (CYP2C8) • ↓ levels with **rifampin, phenytoin**, bile acid sequestrants
Deferiprone (Ferriprox)	**25 mg/kg three times daily** (75 mg/kg/day)	**Boxed warning: agranulocytosis/neutropenia** → **ANC weekly** • **Red-brown urine** • Zinc deficiency, arthralgia

6. Vitamin B12 (Cobalamin) Deficiency

Physiology

- B12 is found only in **animal products** (meat, eggs, dairy) → **vegans** are at risk.
- Absorption needs **stomach acid** (releases B12 from food) + **intrinsic factor (IF)** (from **parietal cells**) → absorbed in the **terminal ileum**.
- Body stores are large (**years**), so deficiency develops slowly (3–5 years).
- **RDA: 2.4 mcg/day**.

Causes

Cause	Examples
Pernicious anemia (most common cause)	**Autoimmune** destruction of parietal cells → **no intrinsic factor** • Antibodies to **intrinsic factor** and parietal cells • Needs **lifelong** B12
Gastric surgery	**Gastrectomy**, **gastric bypass (bariatric)**
Ileal disease	**Crohn's disease**, ileal resection
Diet	**Strict vegans**
Elderly	**Atrophic gastritis** (low acid)
Drugs	**Metformin** (long-term; check B12 periodically), **PPIs**, **H2RAs**, colchicine, **nitrous oxide**
Others	Fish tapeworm, bacterial overgrowth, pancreatic insufficiency

Treatment

Product	Dose	Notes
Cyanocobalamin IM or deep SC	**1,000 mcg daily × 1 week → weekly × 4 weeks → monthly for life** (pernicious anemia)	Preferred for **severe deficiency** or **neurologic symptoms**
Cyanocobalamin oral	**1,000–2,000 mcg daily**	Effective **even in pernicious anemia** (≈1% absorbed by **passive diffusion**, no IF needed) • Cheap
Cyanocobalamin intranasal (Nascobal)	**500 mcg once weekly** in one nostril	For **maintenance** after hematologic remission (not for initial treatment of severe deficiency)
Hydroxocobalamin IM	1,000 mcg (less frequent dosing; longer retention)	Also the **antidote for cyanide poisoning** (Cyanokit)

Key points

- **Never give folic acid alone** to a patient with B12 deficiency: it **corrects the anemia** but **neurologic damage continues** ("**masks**" B12 deficiency).
- **Hypokalemia** can occur in the **first days** of treating severe megaloblastic anemia (potassium moves into new RBCs) → **monitor potassium**.
- **Response:** reticulocytes rise in **3–5 days** (peak ~1 week) • Hgb normalizes in **~2 months** • **Neurologic symptoms** improve over months and **may be permanent** if long-standing.
- **Bariatric surgery and pernicious anemia → lifelong** supplementation.
- ADRs: rare; injection site pain, rare hypersensitivity (cobalt).

7. Folate (Folic Acid) Deficiency

Causes

Cause	Examples
Poor intake	Poor diet, **alcoholism** (most common), elderly
Increased requirements	**Pregnancy**, lactation, **hemolytic anemia** (sickle cell), malignancy
Malabsorption	**Celiac disease**, tropical sprue, IBD
Drugs	**Methotrexate**, **trimethoprim**, **pyrimethamine** (DHFR inhibitors) • **Phenytoin**, phenobarbital, primidone • **Sulfasalazine** • Oral contraceptives (minor)
Dialysis	Folate is removed by dialysis

- Folate stores are small: deficiency develops in **weeks to months** (vs years for B12).
- **RDA: 400 mcg/day** (dietary folate equivalents).

Treatment

Situation	Dose
Folate deficiency	**Folic acid 1 mg PO daily** for ~**4 months** (or until recovery; longer if the cause persists) • 1–5 mg/day in malabsorption
Neural tube defect prevention (all women who could become pregnant)	**400–800 mcg daily**, starting **at least 1 month before conception** and through early pregnancy
Previous pregnancy with neural tube defect (high risk)	**4 mg daily**
Chronic hemolysis (e.g., sickle cell), dialysis	**1 mg daily**
Methotrexate (low-dose, RA/psoriasis)	**Folic acid 1 mg daily** (or 5 mg weekly) to ↓ side effects (mouth sores, GI, liver)

Leucovorin (folinic acid)

- **Active (reduced) form** of folate → **bypasses dihydrofolate reductase (DHFR)**.
- Used for **methotrexate toxicity** ("leucovorin rescue" after high-dose MTX) and with **pyrimethamine** (toxoplasmosis).
- Also used with **5-fluorouracil** (enhances its effect).
- **Folic acid does NOT work** for MTX toxicity because DHFR is blocked.

Key points

- **Always rule out B12 deficiency** before treating with folic acid alone.
- **Folic acid ↓ phenytoin levels** (and phenytoin ↓ folate) → monitor phenytoin levels.
- **Response:** reticulocytosis within ~1 week; Hgb normalizes in ~2 months.

8. Anemia of Chronic Disease / Inflammation

- Seen in **chronic infections**, **autoimmune diseases** (RA, lupus, IBD), **cancer**, heart failure, CKD.
- **Mechanism:** inflammation (IL-6) → ↑ **hepcidin** → iron is **trapped** in macrophages and absorption ↓ ("functional iron deficiency"), plus ↓ EPO response and shorter RBC survival.
- Usually **mild normocytic** (can be microcytic).
- **Labs:** ↓ iron, ↓ **or normal TIBC**, **normal or ↑ ferritin**, ↓ TSAT.
- **Treatment: treat the underlying disease** • Iron only if there is **coexisting absolute iron deficiency** • **ESAs** in selected patients (CKD, chemotherapy) • Transfusion if severe.

9. Anemia of CKD (summary)

- Main cause: **↓ erythropoietin (EPO)** + iron deficiency.
- **Iron first:** give iron if **TSAT ≤ 30% and ferritin ≤ 500 ng/mL • IV iron in hemodialysis**, oral first in non-dialysis.
- **Start ESA when Hgb < 10 g/dL • Do not exceed Hgb 11.5** (FDA: reduce/interrupt if > 11).
- **HIF-PH inhibitors** (daprodustat, vadadustat): oral, **dialysis patients only**.
- See the CKD review for full details.

10. Erythropoiesis-Stimulating Agents (ESAs)

Drug	CKD dose	Chemotherapy-induced anemia dose
Epoetin alfa (Epogen, Procrit, Retacrit)	**50–100 units/kg 3 times weekly** IV/SC	**150 units/kg SC 3 times weekly** or **40,000 units SC weekly**
Darbepoetin alfa (Aranesp)	**0.45 mcg/kg weekly** or **0.75 mcg/kg every 2 weeks**	**2.25 mcg/kg weekly** or **500 mcg every 3 weeks**
Methoxy PEG-epoetin beta (Mircera)	**0.6 mcg/kg every 2 weeks** (or monthly once stable)	**Not indicated** for chemotherapy-induced anemia

Key points

Point	Details
Boxed warnings	↑ Risk of **death, MI, stroke, VTE, vascular access thrombosis** when Hgb > 11 g/dL • **Cancer: shortened survival / tumor progression**
Cancer use	Only for **chemotherapy-induced anemia** with **Hgb < 10** • **Not when the goal of therapy is cure** • **Stop when chemotherapy ends** • Use the lowest dose to avoid transfusion
Contraindications	**Uncontrolled hypertension** • Pure red cell aplasia after ESA
Adverse effects	**Hypertension** (most common), **thrombosis**, seizures, edema, **pure red cell aplasia** (anti-EPO antibodies)
Dose adjustment	**Do not increase more often than every 4 weeks** • ↑ dose 25% if Hgb rises < 1 g/dL after 4 weeks • ↓ dose ≥ 25% if Hgb rises > 1 g/dL in 2 weeks
Iron	Check and replete iron **before and during** therapy (ESAs consume iron)
Hyporesponsiveness	**Iron deficiency (most common)**, infection/inflammation, hyperparathyroidism, aluminum toxicity, B12/folate deficiency, inadequate dialysis
Storage	**Refrigerate**; **do not shake**; do not freeze

11. Hemolytic Anemias

G6PD deficiency

- **X-linked** (mostly males) • common in Middle Eastern, African, Mediterranean, Asian populations.
- RBCs cannot handle **oxidative stress** → **hemolysis** after oxidant drugs, infections, or **fava beans** (favism).
- Smear: **Heinz bodies**, **bite cells**.
- **Treatment: stop the trigger**, supportive care, transfusion if severe.

Drugs to avoid in G6PD deficiency (high-yield):

Class	Drugs
Antimalarials	**Primaquine**, **tafenoquine** (test G6PD before use) • (chloroquine: low risk)
Antibiotics	**Sulfonamides** (TMP-SMX), **dapsone**, **nitrofurantoin**, fluoroquinolones (low risk), chloramphenicol
Others	**Rasburicase** (also methemoglobinemia — contraindicated), **pegloticase**, **methylene blue**, phenazopyridine, sulfonylureas (glyburide; caution), high-dose aspirin, vitamin K (high dose), naphthalene (mothballs)

Drug-induced immune hemolytic anemia

- Drugs: **cephalosporins** (**ceftriaxone**, **cefotetan**: most common), **penicillins** (high dose), **methyldopa** (positive Coombs), **quinine/quinidine**, **NSAIDs**, levodopa, rifampin.
- **DAT (Coombs) positive** • **Treatment: stop the drug** (± steroids).

Autoimmune hemolytic anemia (AIHA)

Type	Antibody	Treatment
Warm AIHA (most common)	**IgG**	**Corticosteroids first-line** (prednisone **1 mg/kg/day**) • **Rituximab** (second-line) • Splenectomy, immunosuppressants (azathioprine, cyclophosphamide)
Cold agglutinin disease	**IgM**	**Avoid cold** • **Rituximab** • **Sutimlimab** (complement C1s inhibitor) • **Steroids are not effective**

Microangiopathic hemolytic anemia (MAHA)

- **Schistocytes** on smear • TTP, HUS, DIC, mechanical heart valves.
- Drug causes: **cyclosporine**, **tacrolimus**, **gemcitabine**, **clopidogrel/ticlopidine**, **quinine**, mitomycin.

- **TTP:** plasma exchange + steroids ± **caplacizumab**, rituximab • **Platelet transfusion generally avoided**.

12. Sickle Cell Disease (SCD)

Basics

- **Autosomal recessive** hemoglobin S (HbSS) → sickled RBCs → **hemolysis** + **vaso-occlusion**.
- Complications: **vaso-occlusive (pain) crisis**, **acute chest syndrome**, **stroke**, splenic sequestration, aplastic crisis (**parvovirus B19**), **functional asplenia** (infections with **encapsulated bacteria**), priapism, avascular necrosis.
- Triggers: **dehydration, infection, cold, hypoxia**, stress, high altitude.

Preventive care

Intervention	Details
Penicillin prophylaxis	**Penicillin VK 125 mg BID** from diagnosis (by 2 months) **until age 3** → **250 mg BID until age 5** • Erythromycin if penicillin-allergic
Vaccines	**Pneumococcal** (PCV + PPSV23), **meningococcal** (ACWY + B), **Hib**, **influenza** yearly, routine vaccines
Folic acid	**1 mg daily** (chronic hemolysis)
Fever ≥ 38.5 °C	**Medical emergency** → urgent evaluation, blood cultures, empiric **IV ceftriaxone**
Stroke screening	**Transcranial Doppler (TCD)** yearly ages **2–16** → abnormal → **chronic transfusions**
Iron overload	From chronic transfusions → **chelation** (deferasirox, deferoxamine, deferiprone)

Disease-modifying therapy

Drug	Dose	Key points
Hydroxyurea (Droxia, Siklos)	Start **15 mg/kg/day** (children 20 mg/kg/day) • ↑ by **5 mg/kg/day every 8 weeks** • Max **35 mg/kg/day**	**First-line** • ↑ **fetal hemoglobin (HbF)** → fewer crises, ACS, transfusions • Offer to all children ≥ 9 months • **Myelosuppression** → **CBC every 2–4 weeks** during titration • **Macrocytosis** expected (adherence marker) • **Teratogenic** → contraception • Avoid **live vaccines** • Nail/skin hyperpigmentation
L-glutamine (Endari)	**0.3 g/kg PO twice daily** (max 15 g/dose), mixed with food/drink	↓ Oxidative stress → fewer crises • GI effects
Crizanlizumab (Adakveo)	**5 mg/kg IV** at week 0 and 2, then **every 4 weeks**	**P-selectin inhibitor** → ↓ vaso-occlusive crises • Infusion reactions • Interferes with platelet count tests
Voxelotor (Oxbryta)	—	**Withdrawn from the market (2024)** due to ↑ deaths and VOCs
Gene therapies	Exagamglogene autotemcel (Casgevy), lovotibeglogene autotemcel (Lyfgenia)	Potentially curative (with HSCT); ages ≥ 12

Acute management

Problem	Management
Vaso-occlusive (pain) crisis	**Hydration** • **Opioids** promptly (morphine, hydromorphone; PCA) • NSAIDs if renal function OK • **Avoid meperidine** (seizures) • O_2 only if hypoxic • Incentive spirometry
Acute chest syndrome	**Antibiotics: ceftriaxone + azithromycin** (cover atypicals) • O_2 • Incentive spirometry • **Transfusion** (simple or exchange) • Analgesia
Acute stroke	**Exchange transfusion**
Aplastic crisis (parvovirus B19)	Transfusion
Splenic sequestration	Transfusion; splenectomy if recurrent

Transfusion target in SCD: avoid Hgb > 10 g/dL with simple transfusion (risk of **hyperviscosity**).

13. Other Anemias

Anemia	Key points
Thalassemia	Microcytic with **normal/high iron**, normal RDW, **target cells**, **Mentzer index < 13** • **Do not give iron** unless iron deficiency is proven • Folic acid • Major forms: **chronic transfusion + iron chelation** • **Luspatercept** (Reblozyl, 1 mg/kg SC every 3 weeks) ↓ transfusion needs (beta-thalassemia, MDS)
Aplastic anemia	**Pancytopenia** + hypocellular marrow • Drug causes: **chloramphenicol**, **carbamazepine**, **felbamate**, **sulfonamides**, gold, NSAIDs, **antithyroid drugs** (methimazole, PTU), zidovudine, chemotherapy, benzene • **Treatment: HSCT** (young) or **ATG + cyclosporine + eltrombopag**
Sideroblastic anemia	Ring sideroblasts • Causes: **isoniazid** (pyridoxine deficiency), **alcohol**, **lead**, **linezolid**, chloramphenicol, copper deficiency/zinc excess • **Treatment:** stop the cause + **pyridoxine (vitamin B6)**
Acute blood loss	Normocytic; Hgb may be **normal initially** (takes hours to equilibrate) • Fluids + transfusion
Drug-induced hemolysis / methemoglobinemia	**Dapsone**, **benzocaine**, **nitrates**, primaquine, **rasburicase** → methemoglobinemia → **methylene blue** (**contraindicated in G6PD deficiency**)

14. Blood Transfusion

- **1 unit of packed RBCs** raises Hgb by about **1 g/dL** (Hct by ~3%).
- **Restrictive threshold:** transfuse if **Hgb < 7 g/dL** in most hospitalized stable patients.
- **Hgb < 8 g/dL**: cardiac surgery, orthopedic surgery, pre-existing cardiovascular disease, symptomatic patients.
- Risks: **transfusion reactions**, **TACO** (circulatory overload), **TRALI**, infection, **iron overload** (chronic), **alloimmunization** (HLA sensitization in transplant candidates).

15. Special Populations

Population	Key points
Pregnancy	Iron needs ↑ (**27 mg/day**) • Prenatal vitamins with iron + **folic acid 400–800 mcg** • IV iron in **2nd/3rd trimester** if oral fails or rapid repletion is needed
Infants and children	**Avoid cow's milk before 12 months** (GI blood loss, low iron) • Iron-fortified formula • Iron supplementation for **breastfed infants from 4 months** (1 mg/kg/day) • Treatment dose **3–6 mg/kg/day**
Elderly	Anemia is **not normal aging** → investigate • B12 deficiency common (atrophic gastritis, metformin, PPIs) • **IDA in elderly → rule out GI cancer**
Bariatric surgery	Lifelong **iron, B12, folate** (and calcium, vitamin D) supplementation

Population	Key points
Vegans	**B12** supplementation required; iron (non-heme iron is less absorbed)
Heart failure	Iron deficiency (ferritin < 100, or 100–299 with TSAT < 20%) → **IV iron** (ferric carboxymaltose/derisomaltose) improves symptoms; **ESAs not recommended**

Exam Traps and Frequently Tested Points

1. **Anemia:** Hgb **< 13** (men), **< 12** (women), **< 11** (pregnancy).
2. **Microcytic** (< 80): **TAILS** • **Macrocytic** (> 100): **B12, folate**, alcohol, drugs.
3. **Ferritin** = best test for iron stores; **low ferritin = iron deficiency**; normal/high in inflammation.
4. **IDA:** ↓ iron, ↑ **TIBC**, ↓ TSAT, ↓ ferritin, ↑ **RDW** • **ACD:** ↓ TIBC, normal/↑ ferritin.
5. **Thalassemia:** microcytic, normal iron studies, normal RDW, **Mentzer < 13 → do not give iron**.
6. **B12 deficiency: ↑ MMA + ↑ homocysteine + neurologic symptoms** • **Folate:** ↑ homocysteine only, normal MMA, no neuro.
7. **Ferrous sulfate 325 mg = 65 mg elemental (20%)** • gluconate 12% • **fumarate 33%**.
8. **Oral iron dose: 60–200 mg elemental/day**; once daily or alternate days is effective with fewer GI effects.
9. Iron: **empty stomach**, **vitamin C ↑**, **avoid antacids, PPIs, calcium, dairy, tea, coffee**.
10. **Iron + tetracyclines/quinolones/levothyroxine/INSTIs** → chelation → **separate**.
11. **Most common iron ADR: constipation**; dark stools are harmless.
12. **Peak reticulocytes 7–10 days**; Hgb ↑ 1–2 g/dL in 2–4 weeks; **continue ~3 months after Hgb normalizes**.
13. **Most common cause of oral iron failure: nonadherence.**
14. **Iron dextran → test dose** (anaphylaxis boxed warning) • **Ferumoxytol → anaphylaxis + MRI interference** • **Ferric carboxymaltose → hypophosphatemia**.
15. **Iron deficit = wt × (target − actual Hgb) × 2.4 + 500**.
16. **Acute iron poisoning → deferoxamine** (vin rosé urine) • **Deferasirox** → renal/hepatic failure, GI bleed • **Deferiprone → agranulocytosis** (ANC weekly).
17. **Pernicious anemia** = no intrinsic factor → **lifelong B12**; **oral high-dose B12 works** (passive diffusion).
18. **B12 IM: 1,000 mcg daily × 1 week → weekly × 4 → monthly for life.**
19. **Never give folic acid alone in B12 deficiency** (masks it; neuro damage continues).
20. **Hypokalemia** early in B12/folate treatment → monitor K.
21. **Metformin, PPIs, H2RAs** → B12 deficiency.
22. **Methotrexate, trimethoprim, pyrimethamine, phenytoin** → folate deficiency.
23. **Folic acid 1 mg daily × 4 months** for deficiency • **400–800 mcg** for NTD prevention • **4 mg** if previous NTD.
24. **Leucovorin** (not folic acid) for **methotrexate toxicity** and with **pyrimethamine**.
25. **Folic acid ↓ phenytoin levels.**
26. **ESAs:** start Hgb < 10; **avoid Hgb > 11**; boxed warning: death, MI, stroke, VTE, tumor progression; **not when cure is the goal**; **stop after chemo ends**.
27. **Most common ESA ADR: hypertension**; **uncontrolled HTN = contraindication**; most common cause of hyporesponse = **iron deficiency**.
28. **G6PD deficiency → avoid** primaquine, tafenoquine, dapsone, sulfonamides, nitrofurantoin, **rasburicase**, **methylene blue**, fava beans.
29. **Ceftriaxone/cefotetan, methyldopa** → immune hemolytic anemia (Coombs positive).
30. **Warm AIHA → steroids** • **Cold agglutinin → avoid cold, rituximab** (steroids don't work).
31. **Sickle cell: hydroxyurea** ↑ HbF (myelosuppression, teratogenic, macrocytosis) • **penicillin prophylaxis until age 5** • **folic acid** • vaccines • **fever = emergency**.
32. **Sickle cell pain crisis:** fluids + **opioids**; **avoid meperidine** • **Acute chest syndrome: ceftriaxone + azithromycin** + transfusion.
33. **Voxelotor** was **withdrawn** (2024).
34. **Isoniazid → sideroblastic anemia → pyridoxine**.

35. **Chloramphenicol, carbamazepine, felbamate** → aplastic anemia.
36. **1 unit PRBC ↑ Hgb ~1 g/dL**; transfuse at **Hgb < 7** (most patients).
37. **IDA in men/postmenopausal women → rule out GI bleeding/cancer.**

> **Note:** Some cut-off values and doses differ slightly between references. **Use the numbers in the reference you are studying for the exam.**

Drug Summary Table: Doses, Key ADRs and DDIs

Drug	Dose	Key ADRs	Key DDIs / special notes
Ferrous sulfate (oral)	325 mg (65 mg elemental) 1–3 × daily, or once daily/alternate days • Children 3–6 mg/kg/day elemental	**Constipation**, nausea, epigastric pain, **dark stools**	↓ by **antacids, PPIs, H2RAs, Ca, dairy, tea, coffee** • Chelates **tetracyclines, quinolones, levothyroxine, INSTIs**, levodopa, bisphosphonates • Vitamin C ↑ absorption • Empty stomach
Ferrous gluconate (oral)	325 mg (≈36 mg elemental) 1–3 × daily	Same as above (less elemental iron/tablet)	Same as ferrous sulfate
Ferrous fumarate (oral)	324 mg (≈106 mg elemental) once daily	Same as above	Same as ferrous sulfate
Polysaccharide-iron / carbonyl iron (oral)	Dose as elemental iron (e.g., 150 mg daily)	Same as above (possibly less GI upset)	Same interactions
Ferric maltol (Accrufer, oral)	30 mg twice daily (empty stomach)	GI effects (milder)	Same interactions
Iron dextran (INFeD, IV)	**Test dose 25 mg**, then calculated total dose	**Anaphylaxis (boxed warning)**, hypotension, arthralgia	**Test dose required** • Stop oral iron
Sodium ferric gluconate (Ferrlecit, IV)	125 mg per HD session × 8 (1 g)	Hypotension, hypersensitivity	No test dose
Iron sucrose (Venofer, IV)	HD: 100 mg × 10 • NDD-CKD: 200 mg × 5 over 14 days	Hypotension, cramps, hypersensitivity	No test dose
Ferumoxytol (Feraheme, IV)	510 mg, then 510 mg 3–8 days later	**Anaphylaxis (boxed warning)**, hypotension	Infuse ≥ 15 min, observe 30 min • **MRI interference** (up to 3 months)
Ferric carboxymaltose (Injectafer, IV)	750 mg × 2 (≥ 7 days apart) • 15 mg/kg if < 50 kg	**Hypophosphatemia**, hypertension, flushing	Monitor phosphate if repeat courses
Ferric derisomaltose (Monoferric, IV)	1,000 mg single dose • 20 mg/kg if < 50 kg	Hypersensitivity	Single-visit repletion
Deferoxamine (Desferal)	Acute poisoning: IV up to 15 mg/kg/h • Chronic: 20–60 mg/kg/day SC over 8–12 h	Hypotension (rapid IV), **ototoxicity, retinal toxicity**, injection site reactions, **Yersinia** infection	**Vin rosé urine** • Avoid vitamin C > 200 mg/day early (cardiac toxicity)
Deferasirox (Exjade / Jadenu)	Exjade 20 mg/kg/day (empty stomach) • Jadenu 14 mg/kg/day	**Boxed: renal failure, hepatic failure, GI hemorrhage** • Rash, GI upset	Avoid **aluminum antacids** • ↑ **repaglinide** (CYP2C8) • ↓ by **rifampin, phenytoin, cholestyramine**
Deferiprone (Ferriprox)	25 mg/kg TID (75 mg/kg/day)	**Boxed: agranulocytosis/ neutropenia** • arthralgia, zinc deficiency	**ANC weekly** • **Red-brown urine** • Separate from polyvalent cations (Fe, Al, Zn) by 4 h

Drug	Dose	Key ADRs	Key DDIs / special notes
Cyanocobalamin IM/SC	1,000 mcg daily × 1 wk → weekly × 4 wk → monthly for life	Injection site pain; rare anaphylaxis; **hypokalemia** early in treatment	Monitor K • Lifelong in pernicious anemia/bariatric surgery
Cyanocobalamin oral	1,000–2,000 mcg daily	Rare	Works even in pernicious anemia (passive diffusion) • Metformin, PPIs, H2RAs ↓ B12 absorption
Cyanocobalamin intranasal (Nascobal)	500 mcg once weekly	Nasal irritation, headache	Maintenance only (after remission)
Hydroxocobalamin IM	1,000 mcg (less frequent dosing than cyanocobalamin)	Injection site pain; red urine/skin with high (Cyanokit) doses	Also cyanide antidote
Folic acid	Deficiency: 1 mg daily × ~4 months • NTD prevention: 0.4–0.8 mg daily • Previous NTD: 4 mg daily • With MTX: 1 mg daily	Rare (well tolerated)	**↓ Phenytoin levels • Rule out B12 deficiency first** (masks it) • MTX, trimethoprim, pyrimethamine, phenytoin, sulfasalazine ↓ folate
Leucovorin (folinic acid)	Varies (MTX rescue; with pyrimethamine 10–25 mg daily)	Rare (allergic reactions)	Bypasses DHFR → antidote for **MTX toxicity** • ↑ 5-FU toxicity • Do not give intrathecally
Epoetin alfa (Epogen, Procrit, Retacrit)	CKD: 50–100 units/kg 3 × weekly • Chemo: 150 units/kg 3 × weekly or 40,000 units weekly	**Hypertension**, thrombosis, seizures, PRCA • **Boxed: death, MI, stroke, VTE, tumor progression**	Avoid Hgb > 11 • Replete iron • Contraindicated in uncontrolled HTN • Refrigerate, don't shake
Darbepoetin alfa (Aranesp)	CKD: 0.45 mcg/kg weekly or 0.75 mcg/kg q2w • Chemo: 2.25 mcg/kg weekly or 500 mcg q3w	Same as epoetin	Longer half-life → less frequent dosing
Methoxy PEG-epoetin beta (Mircera)	CKD: 0.6 mcg/kg q2w (monthly once stable)	Same as epoetin	**CKD only** (not for chemotherapy-induced anemia)
Daprodustat (Jesduvroq)	Oral daily (dose by Hgb and prior ESA dose)	**Boxed: death, MI, stroke, VTE** • HTN	**Dialysis patients only** • **Contraindicated with gemfibrozil** (strong CYP2C8 inhibitor)
Vadadustat (Vafseo)	Oral daily	**Boxed: death, MI, stroke, VTE** • hepatotoxicity, diarrhea	**Dialysis patients only** • Separate from iron and phosphate binders
Hydroxyurea (Droxia, Siklos)	15 mg/kg/day (children 20) → ↑ 5 mg/kg/day q8 wk → max 35 mg/kg/day	**Myelosuppression**, **macrocytosis**, nail/skin hyperpigmentation, GI • **Teratogenic**	CBC every 2–4 wk during titration • Contraception • Avoid **live vaccines** • Caution with other myelosuppressants and antiretrovirals (didanosine, stavudine: pancreatitis)
L-glutamine (Endari)	0.3 g/kg twice daily (max 15 g/dose)	Constipation, nausea, headache, abdominal pain	Mix with food or drink
Crizanlizumab (Adakveo)	5 mg/kg IV wk 0, 2, then q4w	Infusion reactions, arthralgia, nausea, fever	Interferes with automated platelet counts
Penicillin VK (SCD prophylaxis)	125 mg BID until age 3 → 250 mg BID until age 5	Rash, GI	Erythromycin if penicillin-allergic
Prednisone (warm AIHA)	1 mg/kg/day, then taper	Hyperglycemia, infection, osteoporosis, mood changes	Not effective in cold agglutinin disease
Rituximab (AIHA)	375 mg/m² IV weekly × 4 (common regimen)	Infusion reactions • **Boxed: HBV reactivation, PML**	Screen for **hepatitis B** before use

Drug	Dose	Key ADRs	Key DDIs / special notes
Luspatercept (Reblozyl)	1 mg/kg SC every 3 weeks (beta-thalassemia)	Thrombosis, hypertension, bone pain	↓ Transfusion needs (thalassemia, MDS)
Pyridoxine (vitamin B6)	e.g., 50–200 mg/day for sideroblastic anemia; 25–50 mg/day with isoniazid (prevention)	Neuropathy at very high doses	Prevents/treats **isoniazid**-induced neuropathy and sideroblastic anemia

Good luck on your exam!